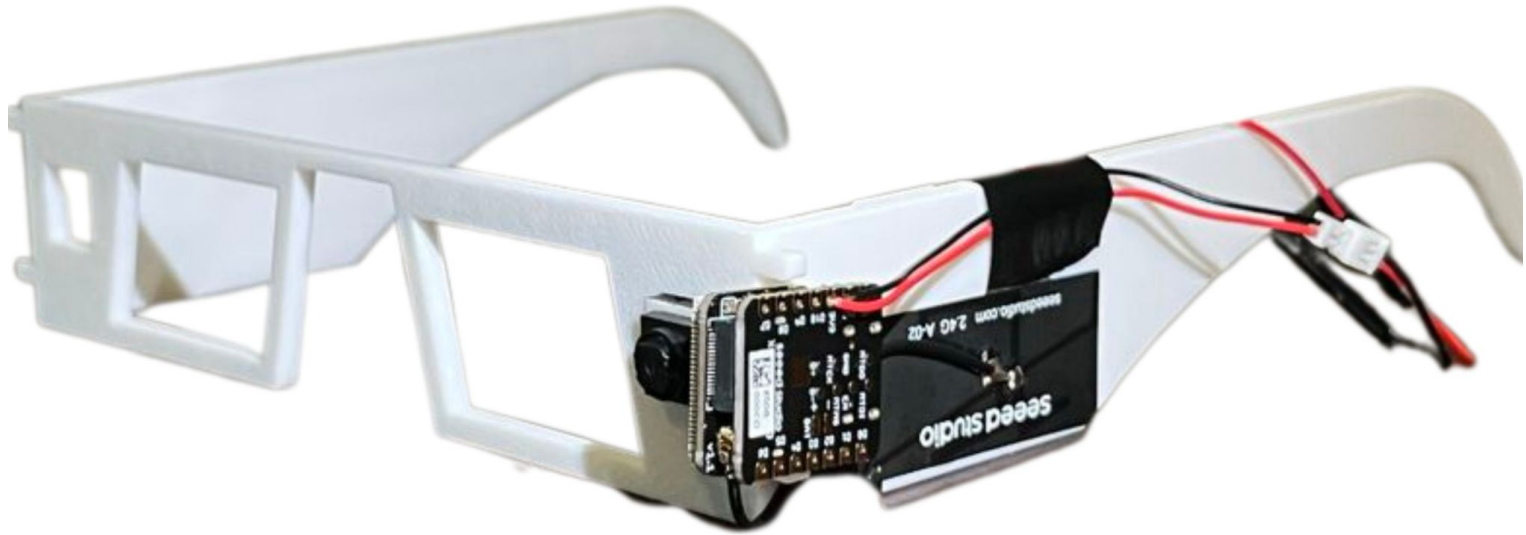




SightLine: AI Powered
Navigation for the Visually Impaired

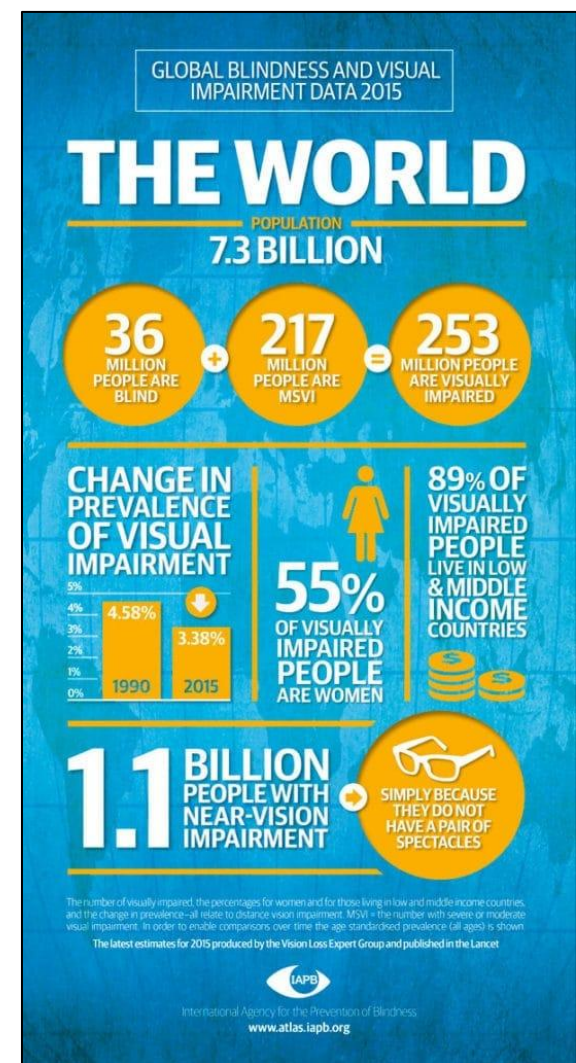
What is SightLine?



#1: Problem Statement

Problem/Target Audience

- In 2020, National Health Institute estimated 253 million individuals suffered from visual impairment globally [1].
- Low-middle income countries are disproportionately affected. [2].
- Papers have expressed modern concerns about rapid AI advancement neglecting individuals with disabilities [3]
- **Time Critical:** ~10 million increase in severe visual impairment between 2015-2020 and AI prices are unaffordable in developing countries. [1, 2]



Impact:

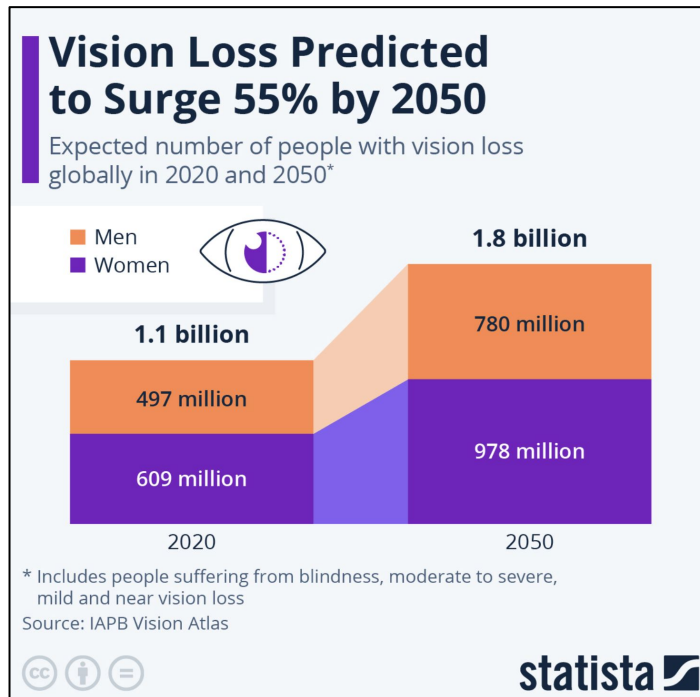
UN Sustainable Development Goal (SDG) #10:
Reduce inequality within and among countries

Method #1:

Reduce inaccessibility to convenient navigation for visually impaired

Method #2:

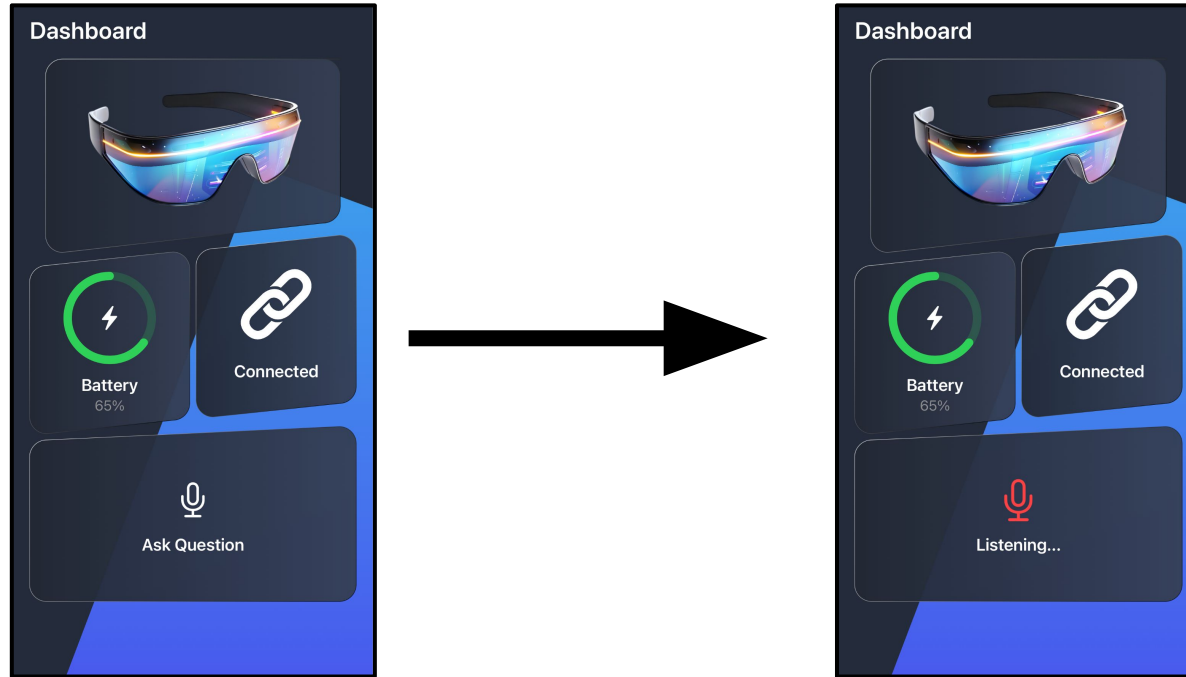
Architect AI system in a cheap, modular way, such that developing countries aren't left behind



Product:	Envision Glasses	RayBanz (Limited Capabilities)	Ours
Price:	\$2,499	\$299	~\$30 (Hardware Cost)

#2: Artificial Intelligence Usage/System/Integration

User Experience (Privacy & Accessibility Oriented)



Voice Activation →
Available to all users

Model Fine-Tuning/Data Collection:

#1: Compiled open-source dataset of indoor furniture from Roboflow

#2 (Considered Target Audience!): Used 1500+ images from diverse indoor settings to account for cultural differences.

#3: Applied augmentation (noise) to increase dataset size (manipulates pixels in images)

#4: Tuned **Pytorch-based** YOLOv8 object detector on dataset, to recognize key objects such as doorways.

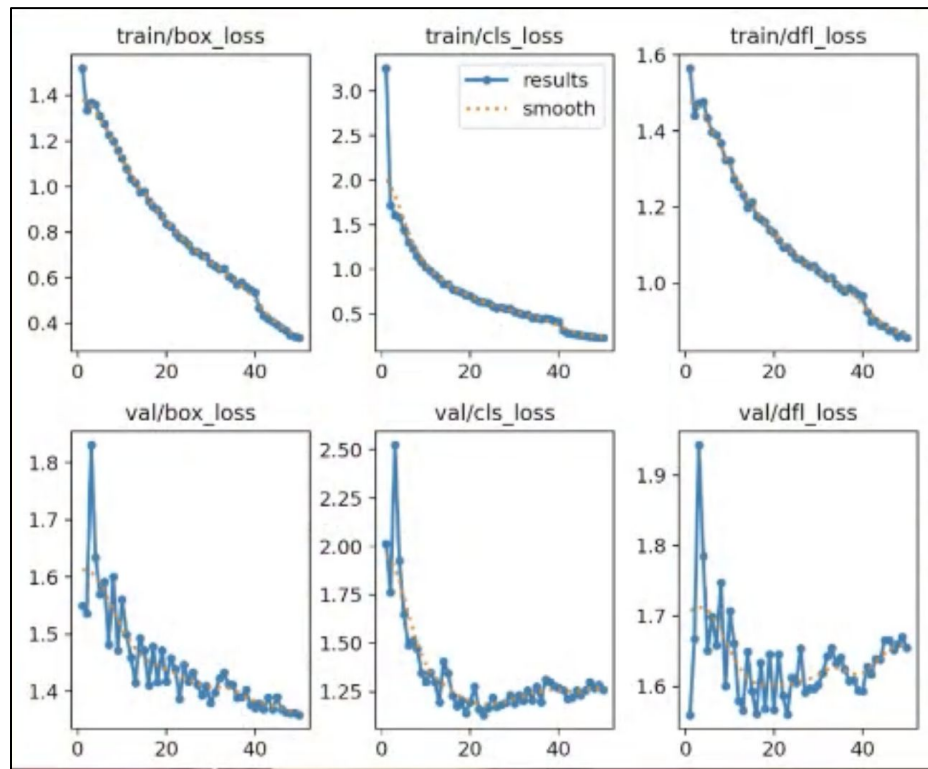
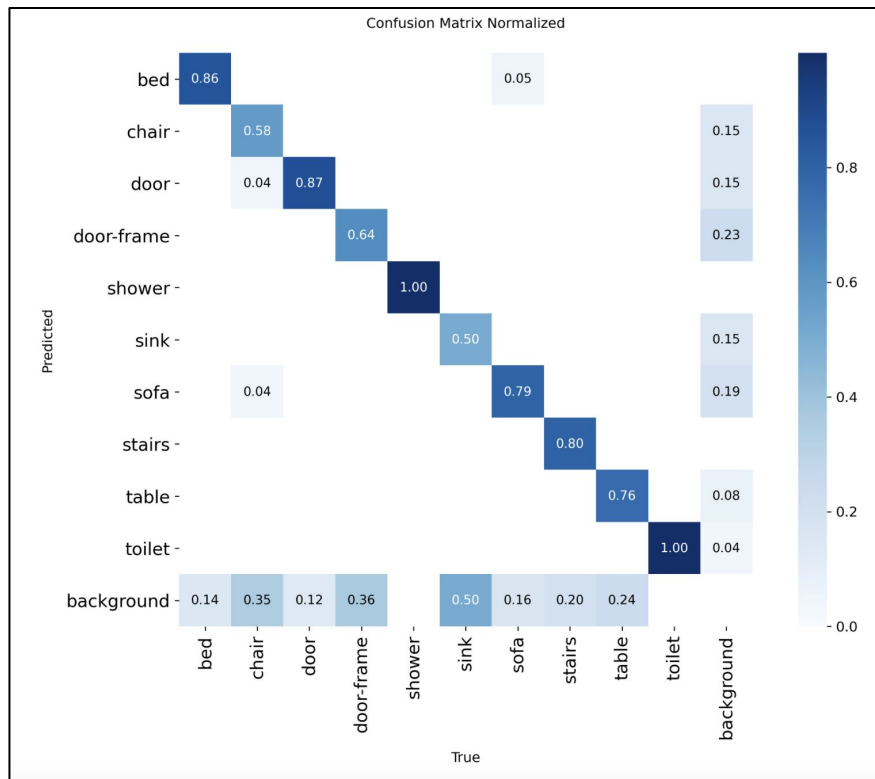


Original

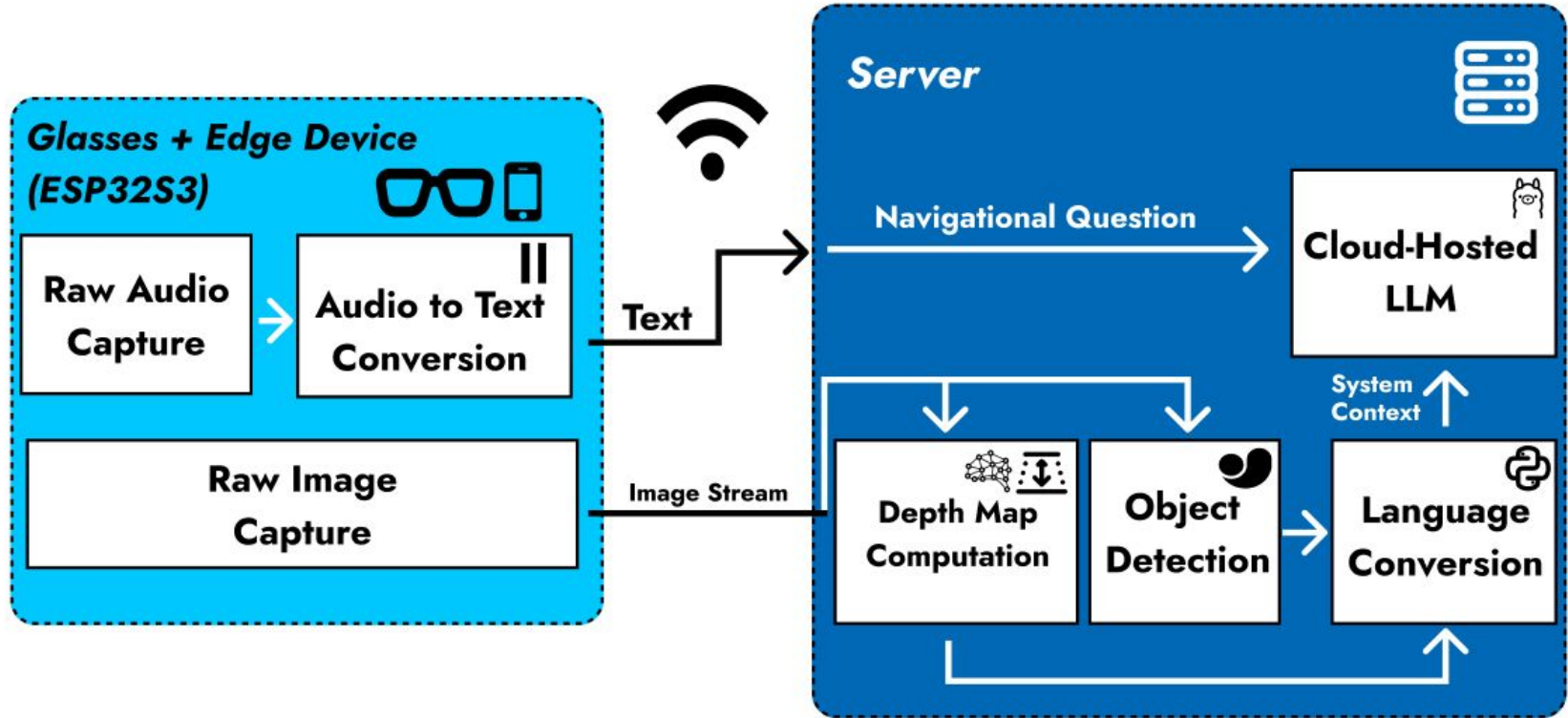


Augmented

Training Results:

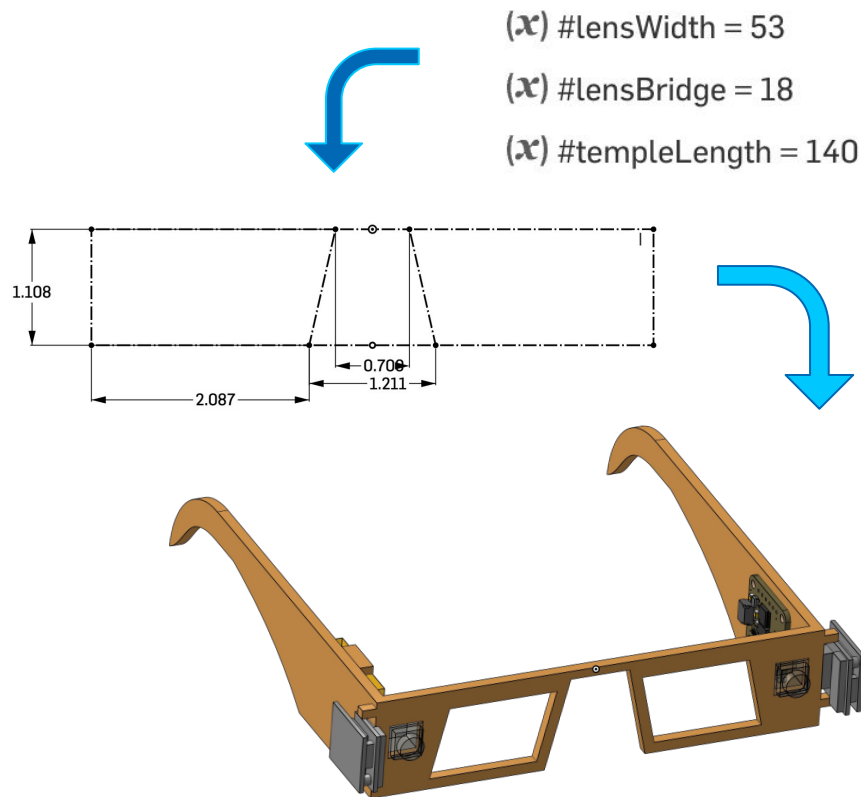


System Architecture:



Hardware

- Lightweight for everyday use ~ 40g
- Quick to assemble, intuitive to learn, effortless to use
- One solid piece, zero bolts or screws
- Easily adjustable to various user face structures



#3: Ethical Considerations

Ethical Consideration #1:

Privacy, Security, Safety

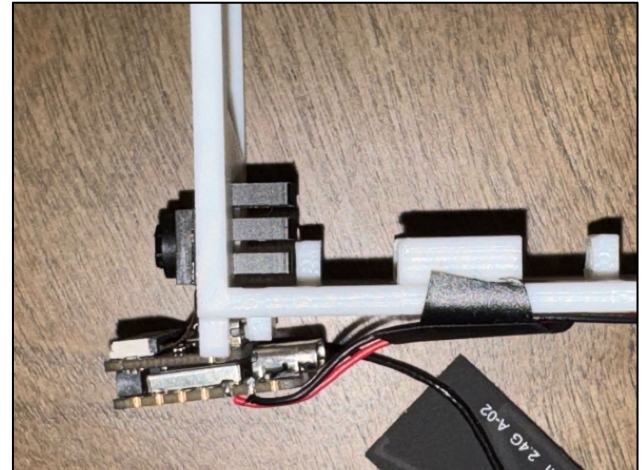
Software:

- All voice data **never** leaves the device, once processing is finished it is deleted
- Images are **never** permanently stored on the server and are deleted after processing

Hardware:

- Heat sink component to prevent overheating injury/fire hazards

```
Based on the visual analysis of the environment, the following objects have been detected:  
- A chair located at approximate coordinates (1662, 1856) with an estimated depth of 3.48 meters.  
- A chair located at approximate coordinates (1029, 1404) with an estimated depth of 6.42 meters.  
- A chair located at approximate coordinates (2549, 1720) with an estimated depth of 4.23 meters.  
- A dining table located at approximate coordinates (2048, 1752) with an estimated depth of 5.55 meters.  
- A dining table located at approximate coordinates (1511, 1484) with an estimated depth of 5.64 meters.  
- A dining table located at approximate coordinates (1481, 1376) with an estimated depth of 4.99 meters.
```

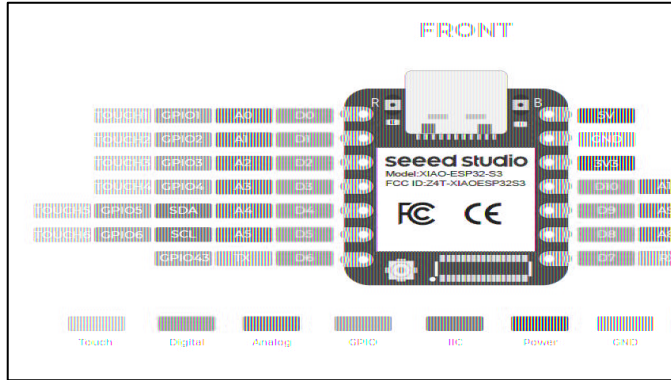


Ethical Consideration #2:

Bias, Discrimination: Trained model on diverse context images + affordable API via Intel Hosting



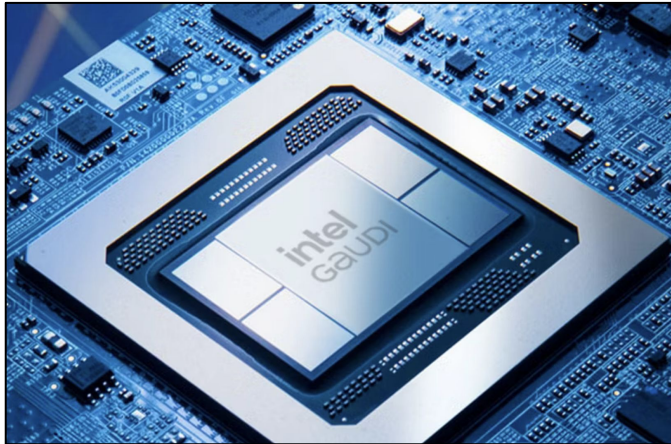
Ethical Consideration #3:



Environmental Sustainability:

ESP32S3 is a small and power efficient system

Intel Gaudi accelerator has greater power efficiency for model inference over competitors (Deployment)

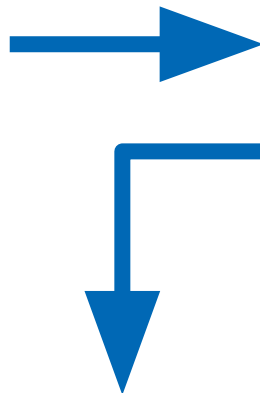


#4: Intel Technologies & Scalability Plan

Intel Technologies Project Lifecycle

Prototyping:

- 13th Gen Intel Core i7 CPU (Inference Testing)
- OpenVINO Jupyter Notebooks (OpenVINO Example Code)



Artificial Intelligence Optimization:

- OpenVINO Toolkit and NNCF for post-training INT8 quantization (~2-4x faster)

Scaling/Deployment Plan:

- Habana Gaudi accelerators for CV training (address more blind user tasks) AND persistent model hosting
- Tiber AI Cloud for deploying on a Kubernetes cluster (hosts model pipeline) allowing API creation + connection to glasses + easily scalable.

Go-To-Market/Scale

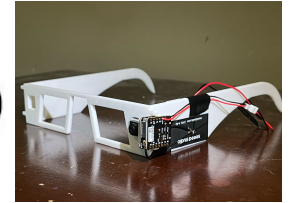
Step #1: Deploy current pipeline on Intel DevCloud and connect to web testing interface



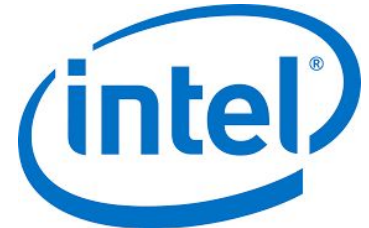
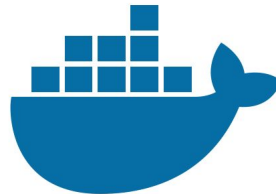
Step #2: Collaborate with Intel mentors to adapt Depth Pro deep learning algorithm to be compatible with Intel Hardware.



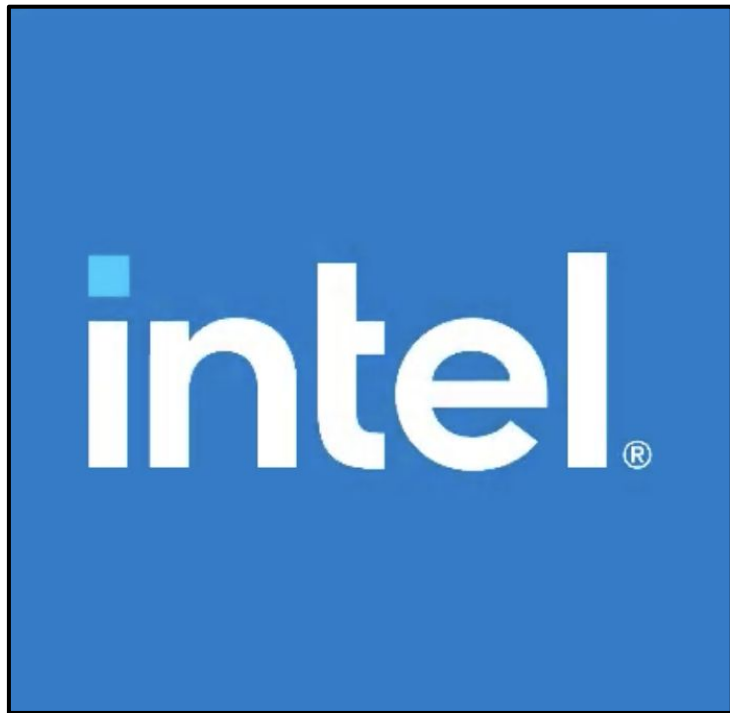
Step #3: Convert Intel-hosted pipeline into public API and license to glasses providers / make own glasses pair.



Tools for Model Scaling:



THANK YOU



X



References:

- [1] Ackland, Peter, et al. "World Blindness and Visual Impairment: Despite Many Successes, the Problem Is Growing." Community Eye Health, vol. 30, no. 100, 2017, pp. 71–73. PubMed Central, <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5820628/>.
- [2] Outreach, R. (2018, July 30). IAPB: Envisioning the future of universal eye care. Research Outreach. <https://researchoutreach.org/articles/iapb-envisioning-the-future-of-universal-eye-care/>
- [3] Guo, Anhong, et al. Toward Fairness in AI for People with Disabilities: A Research Roadmap. arXiv:1907.02227, arXiv, 2 Aug. 2019. arXiv.org, <https://doi.org/10.48550/arXiv.1907.02227>.
- [4] Gamage, B., Do, T.-T., Price, N. S. C., Lowery, A., & Marriott, K. (2023). What do blind and low-vision people really want from assistive smart devices? Comparison of the literature with a focus study. The 25th International ACM SIGACCESS Conference on Computers and Accessibility, 1–21. <https://doi.org/10.1145/3597638.3608955>
- [5] Bochkovskii, Aleksei, et al. Depth Pro: Sharp Monocular Metric Depth in Less Than a Second. arXiv:2410.02073, arXiv, 21 Apr. 2025. arXiv.org, <https://doi.org/10.48550/arXiv.2410.02073>.
- [6] *Infographic: Vision loss predicted to surge 55% by 2050*. (2024, January 4). Statista Daily Data. <https://www.statista.com/chart/31502/expected-number-of-people-with-vision-loss-globally>

OpenVINO Statistics (Reference):

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Metric		Standard YOLOv8	OpenVINO INT8

Disk Size (MB)		21.48	21.84
RAM Usage (MB)		61.60	148.02

Latency (ms/img)		266.50	90.49
Throughput (FPS)		3.75	11.05

mAP @50		0.7759	0.5732
F1-Score		0.7629	0.5966
Precision		0.7762	0.7889
Recall		0.7500	0.4797